SCOREBOARD – Hadamard 14072 Paley-I (2026-10-08)

Cutoff: 2026-10-08T09:30:39Z (~5:30AM ET)

Object	Status	max|HHˆT-nI|	Wall (s)	SPK
H_14072 Paley-I (q=14071)	PASS	0	2519.4855	**IN SPK** (Paley-I)
H_14072 Paley-II (q=7035)	FAIL/SKIP	–	–	q=7035=3*57*67 NOT prime; ===3 mod 4 needs ===1
H_14072 Kronecker (4*3518)	FAIL/SKIP	–	–	3518===2 mod 4 NOT Hadamard-admissible; factors 2*1759

Units: 642/642 * row blocks 110/110 * overall_pass=True * wall_total=2560.0771s
Gap 2052: **OPEN** * clay_closed=false * GATE-04 **LOCKED** * conjecture **OPEN**
IN SPK via Paley-I; Miyamoto digression skipped; exclude Miyamoto-only 8948/8644/8516/...; prior SPK 10224+10212+10164+10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884 retained

Next optional: **q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3ˆ271*317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59ˆ2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2ˆ2*59ˆ2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2ˆ2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3ˆ2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2ˆ2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1367 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2ˆ2*3*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13ˆ2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*1167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2ˆ2*3*1167 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | FIRST Paley-II DIRECT after 14072: q2=7057->n=14116 (q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5*941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2ˆ2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII); Paley-I DIRECT 14084/14088/14108 come first) | Kronecker-only (no direct Paley route; note-only, NOT on box): none in 14072..14084 (14080=4*3520, 3520=2?*511 ===0 mod 4 order-admissible but H_3520 has no direct classical Paley route: 3519=3ˆ2*1723 composite; 1759 prime but ===3 mod 4); beyond: 14112=4*3528 (3528=2^(3^ˆ(2^7))*2 ===0 mod 4; H_3528**

*via Paley-I q=3527 prime ===3 mod 4) | next Paley-I DIRECT: q=14083->n=14084 (prime ===3 mod 4; 14081 prime ===1 mod 4 skipped; no prime ===3 mod 4 in 14072..14082; FIRST n===0 mod 4 after 14072 with any direct classical Paley route (14076/14080 none); Paley-II q=7041=3*2347 NOT prime (===1 mod 4 but composite) -> FAIL/SKIP; Kronecker 14084=4*3521; 3521=7503 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3521); 14084=2^27503 ===4 mod 16 = 14052/13908/13860-like odd-m shape (q2 composite ===1, m===1 mod 4)) (or gap 2052 / Track B)*